Name _____
Student Number _____

Tutorial 1 Worksheet Answer guide

I- Are viruses living organisms? Add a checkmark or an X in the following table to compare viruses to prokaryotic and eukaryotic cells.

Features	Viruses	Prokaryotic cells	Eukaryotic cells
Genetic information	✓	✓	✓
Proteins	✓	✓	✓
Nucleus	**X**	**X** **Nucleoid**	✓
Membrane/Envelope	✓ For enveloped virus	✓ Plasma	✓ Plasma Nuclear
Replication Machinery	X/✓ (some do)	✓	✓
Independent metabolism	**X**	✓	✓
Homeostasis	**X**	✓	✓
Ability to evolve	✓	✓	✓

II- What is the difference between a virion and a virus?

Virion: refers to a singular structural unit of a virus, a viral particle.
Virus: more of a general term referring to a biological entity that can be transmitted from cell to cell or host to host. The term virus is also used more generally when referring to the classification of such entities.

Here is a handy reference:

https://www.virology.ws/2004/07/28/what-is-a-virus/

III- List some disadvantages and advantages of viruses (Note that the question asks "of" viruses but not "to" viruses)

Disadvantages	Advantages
- They cause diseases - They have impact on economic system, society, and politics - Commensal viruses can predispose to diseases	- They allow us to better understand cell and molecular biology (RNA polymerase for PCR) - They can be used as therapeutics: used as vectors to insert genes into cells or make them express certain proteins (gene therapies, vaccines, cancer therapies) - Commensal viruses can protect from other diseases - Potential to repurpose antivirals against a specific virus for other viruses with similar life cycles

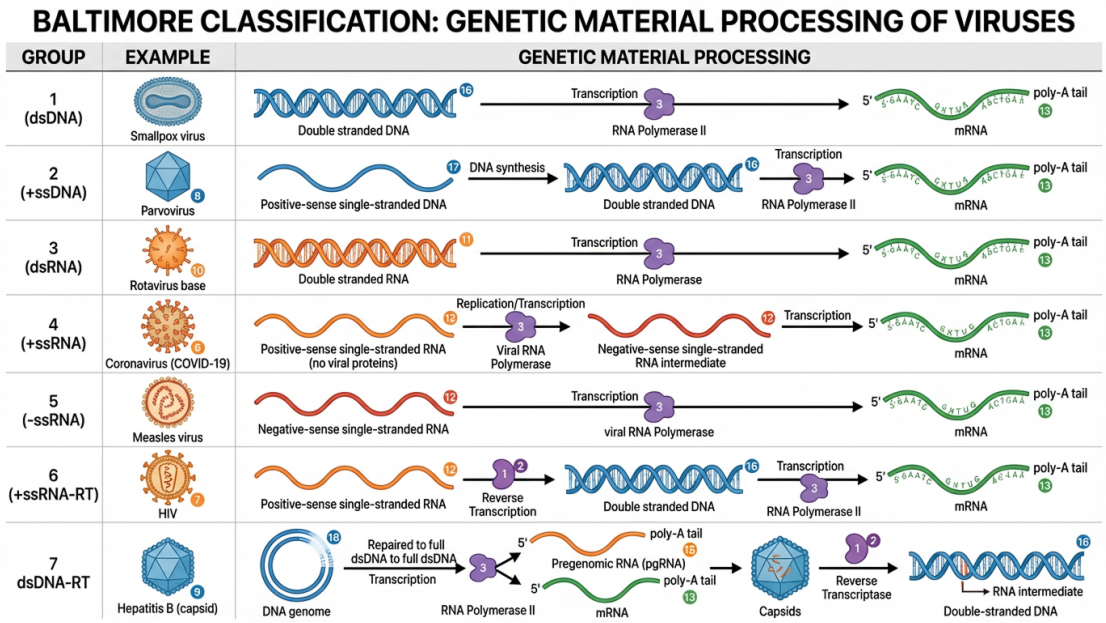
IV- The Baltimore classification scheme is based on the relationship between the virus' genome and its mRNA. Next to each class of virus, indicate where the virus would generally insert its genomes once inside the host cell.

Class	Location of genome Insertion
Class I: ds DNA	nucleus/nucleoid
Class II: ss DNA	nucleus/nucleoid
Class III: ds RNA	cytoplasm
Class IV: +ssRNA	cytoplasm
Class V: - ssRNA	cytoplasm
Class VI: Retroviruses	nucleus/nucleoid (reverse transcription; need the nucleus/nucleoid)
Class VII: dsDNA with an RNA intermediate	nucleus/nucleoid

V- Which of the classes of viruses would need to encode an RNA dependent RNA polymerase?

C. Classes III, IV, V: RNA-dependent RNA polymerase (RdRp) is an essential protein encoded in the genomes of all RNA-containing viruses with no DNA stage i.e., only RNA viruses. It catalyzes synthesis of the RNA strand complementary to a given RNA template.

Here is a handy reference:



VI- When virus capsids are assembled from the protein subunits, they form a type of structure that is described as "metastable".

a) What does it mean to be a metastable structure?
The virus particle has to be stable enough to withstand the extracellular environment but unstable enough to come apart when releasing the genome into the cell.

b) What type(s) of molecular interactions would you expect to observe between the subunits of the capsid?
Would expect to see non-covalent interactions ex: ionic interactions, hydrogen bonds, induced dipole interactions (ie: Hydrophobic effect). Would not expect covalent bonds.

c) What structural aspects of a virus would affect stability?
Is it enveloped or not? Phospholipid bilayer sensitive to -- detergents (amphipathic) – desiccation – Heat -- pH

d) How would the stability of viruses affect work in a laboratory?
Sanitization, Integrity of samples

VII- What is the difference between a (+) sense RNA virus and a (-) sense ssRNA virus? Can you give an example of a type of each virus?

(+) sense ssRNA virus	(-) sense ssRNA virus
(+) sense RNA virus genomes can be translated directly into protein e.g., flavivirus, coronavirus, alphavirus	(-) sense RNA genomes must package an RNA dependent RNA polymerase within their virion, which allows the virus to synthesize (+) RNA, which can then be translated into protein e.g., influenza, pneumovirus, orthoebolavirus